

Sheet 6

$$1. \quad x_t = \alpha_0 + \alpha_1 x_{t-1} + \varepsilon_t \quad (1)$$

$$\alpha_0 = 0.5 \quad \alpha_1 = 0.8, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2), \quad \sigma^2 = 4$$

$$E[x_0] = 2.5 = \mu_0 \quad \text{Var}(x_0) = 11.11 = \sigma_0^2$$

a) We can solve (1) by backwards substitution. Note that

$$x_t = \alpha_0 + \alpha_1 (\alpha_0 + \alpha_1 x_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \quad [k=2]$$

$$= \alpha_0 + \alpha_1 (\alpha_0 + \alpha_1 (\alpha_0 + \alpha_1 x_{t-3} + \varepsilon_{t-2}) + \varepsilon_{t-1}) + \varepsilon_t \quad [k=3]$$

...

$$= \alpha_0 + \alpha_1^t x_0 + \sum_{i=0}^{t-1} \varepsilon_{t-i} \alpha_1^i \quad [k=t]$$

$$\text{or, more generally, } x_t = \alpha_0 \sum_{i=0}^{k-1} \alpha_1^i + \alpha_1^k x_{t-k} + \sum_{i=0}^{k-1} \varepsilon_{t-i} \alpha_1^i$$

Letting $k \rightarrow \infty$, since $|\alpha_1| < 1$, we have $\alpha_1^k \rightarrow 0$; $\sum_{i=0}^{k-1} \alpha_1^i \rightarrow \frac{1}{1-\alpha_1}$

1a. yes, correct!

$$\therefore x_t = \frac{\alpha_0}{1-\alpha_1} + \sum_{i=0}^{\infty} \varepsilon_{t-i} \alpha_1^i$$

(You could also show this with lag polynomials)

$$= 2.5 + \sum_{i=0}^{\infty} \varepsilon_{t-i} \cdot 0.8^i$$

$$b) \quad E[x_t] = \alpha_0 + \alpha_1 E[x_{t-1}] + E[\varepsilon_t] \quad \text{by linearity}$$

when $t=2$, we have

$$E[x_2] = 0.5 + 0.8 \times 2.5 + 0 = 2.5 = E[x_0]$$

so rolling forward / by induction we can see that $\forall t, E[x_t] = 2.5$.

Alternatively, note that

$$E[x_t] = \alpha_0 \sum_{i=0}^{t-1} \alpha_1^i + \alpha_1^t E[x_0] + \sum_{i=0}^{t-1} \alpha_1^i E[\varepsilon_{t-i}]$$

$$= 0.5 \cdot \frac{1-0.8^t}{0.2} + 0.8^t \cdot 2.5$$

$$= 2.5 (1-0.8^t + 0.8^t) = 2.5$$

$$\text{Var}(x_t) = \text{var}(\alpha_1^t x_0 + \sum_{i=0}^{t-1} \varepsilon_{t-i} \alpha_1^i) = \alpha_1^{2t} \text{var}(x_0) + \sum_{i=0}^{t-1} \alpha_1^{2i} \text{var}(\varepsilon_{t-i})$$

since $\varepsilon_i \perp \varepsilon_j$ $i \neq j$ and assuming $\{\varepsilon_t\} \perp x_0$

$$= \alpha_1^{2t} \text{var}(x_0) + \sigma^2 \cdot \frac{1-\alpha_1^{2t}}{1-\alpha_1^2} \quad \text{by sum of series and } \varepsilon_t \text{ ident. distr.}$$

$$= 0.8^{2t} \cdot 11.11 + 4 \cdot \frac{1-0.9^{2t}}{1-0.64} = 11.11 (0.9^{2t} + \frac{1-0.9^{2t}}{1-0.64}) = 11.11 = \text{var}(x_0)$$

1b. yes, all correct! for the autocovariance and correlation, a succinct way of covering both -ve and +ve values of h would be to write $|h|$ i.e., absolute value of h instead of just h in the final

We can define $y_t = x_t - \mu$ where $\mu = E[x_t] = 2.5$
 Then $\gamma_x(h) = E[y_t y_{t-h}]$

$$= E\left[\left(\alpha_1^t y_0 + \sum_{i=0}^{t-1} \alpha_1^i \varepsilon_{t-i}\right) \left(\alpha_1^{t-h} y_0 + \sum_{i=0}^{t-h-1} \alpha_1^i \varepsilon_{t-h-i}\right)\right]$$

$$= E\left[\alpha_1^{2t-h} y_0^2\right] + A \quad \text{where } A := E\left[\sum_{i=0}^{t-1} \alpha_1^i \varepsilon_{t-i} \left(\sum_{j=0}^{t-h-1} \alpha_1^j \varepsilon_{t-h-j}\right)\right]$$

Since y_0 uncorrelated with $\{\varepsilon_t\}$

$$\text{and } A = E\left[\sum_{i=0}^{t-1} \sum_{j=0}^{t-h-1} \alpha_1^i \varepsilon_{t-i} \alpha_1^j \varepsilon_{t-h-j}\right] = \sum_{i=0}^{t-1} \sum_{j=0}^{t-h-1} \alpha_1^{i+j} E[\varepsilon_{t-i} \varepsilon_{t-h-j}] \quad \text{by linearity.}$$

Since errors are uncorrelated, $E[\varepsilon_{t-i} \varepsilon_{t-h-j}] \neq 0$ only when $i=h+j$,
 and in this case $E[\cdot] = \sigma^2$.

$$\text{So } A = \sum_{j=0}^{t-h-1} \alpha_1^{h+2j} \sigma^2 = \alpha_1^h \sigma^2 \sum_{j=0}^{t-h-1} \alpha_1^{2j} = \alpha_1^h \sigma^2 \frac{1 - \alpha_1^{2(t-h)}}{1 - \alpha_1^2}$$

Therefore $\gamma_x(h) = 11.1 \alpha_1^{2t-h} + \alpha_1^h (1 - \alpha_1^{2(t-h)}) \cdot 4/0.36$ as

$$= 11.1 (\alpha_1^{2t-h} + \alpha_1^h - \alpha_1^{2t-h}) \quad \text{var}(y_0) = \text{var}(x_0)$$

$$= 11.1 \alpha_1^h$$

$$\rho_x(h) = \gamma_x(h) / \sqrt{\text{var}(x_t) \text{var}(x_{t-h})} = \gamma_x(h) / 11.1 = \alpha_1^h$$

Since autocovariance is symmetric, we can compute $\gamma_x(h)$ for negative values easily. The autocovariance declines exponentially with lag distance h , starting from $\gamma_x(0) = \text{var}(x_0) = 11.1$.

not sure I follow the reasoning here...(note that we are talking about the sample mean of the process x_t and not the OLS estimator of α_0, α_1 etc)

c) $\bar{x}_{100} := \frac{1}{100} \sum_{i=1}^{100} x_i$. Notice that, assuming $x_0 \sim N(2.5, 11.1)$, x_t is a linear combination of Normal random variables for all t , since the errors are normal.

So, $\bar{x}_{100}$ will also be Normal. Given this, we can perform exact inference (otherwise, we'd need to apply the CLT to show to get the limiting distribution).

1c. unfortunately, this is incorrect.

Observe that the sample mean $\bar{x}_T$ is an unbiased estimator of the mean - if $E[\bar{x}_T] = E[\frac{1}{T} \sum x_t] = \frac{1}{T} \sum E[x_t] = \frac{1}{T} \mu T = \mu$ by linearity and the fact that the process (1) is stationary, (as we showed above) so $E[x_t] = \mu$.

(Note that if $\alpha_0 \neq 0.5$ then the initial conditions specified will not generate a stationary process. But I think the question is assuming the process has run for long enough that by the time $t=1$, it's roughly stationary despite any initial disturbance.)

$$\begin{aligned} \text{var}(\bar{x}_T) &= \frac{1}{T^2} \text{var}\left(\sum_{t=1}^T x_t\right) = \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \text{cov}(x_s, x_t) \\ &= \frac{1}{T^2} \left[\sum_{t=1}^T \text{var}(x_t) + 2 \sum_{t=1}^{T-1} \sum_{s=t+1}^T \text{cov}(x_s, x_t) \right] \\ &= \frac{1}{T} \text{var}(x_t) + C \end{aligned}$$

as stationary $\text{cov}(t)$ depends only on lag $h = s - t$, since stationary.

Analysing C , if $T=1$ is $2 \sum_{h=1}^{T-1} (T-h) r_x(h)$, since there are $2(T-h)$ pairs of observations h periods apart.

$$\therefore \text{var}(\bar{x}_T) = \frac{1}{T} \text{var}(x_t) + 2/T^2 \sum_{h=1}^{T-1} (T-h) r_x(h)$$

We already found $\text{var}(x_t)$ and $r_x(h)$; these are unaffected by the constant term α_0 .

$$\text{So } \text{var}(\bar{x}_{100}) = 0.1111 + \frac{1}{5000} \sum_{h=1}^{99} (100-h) \cdot 11.11 \cdot 0.8^h$$

Since $r_x(h)$ falls exponentially with h , the ~~terms~~ ^{elements} in the second term will become negligible quickly. We have $\frac{1}{5000} \cdot (979 + 696 + 551 + 436 + 345 + 273 + 216 + 171)$ computing the first 8 terms, and can conclude $\text{var}(\bar{x}_{100}) \approx 0.82$.

So $\bar{x}_{100} \sim N(\mu, 0.82)$; ~~then~~ ^{now} the 95% CI is constructed with $C_\alpha = 1.96$ centred around $\bar{x}_{100}^{\text{obs}} = 0.03$, i.e. the range $[-1.58, 1.64]$.

Maybe I should * actually calculate $\sum_{h=1}^{99} (100-h) \cdot 0.8^h$ but it's complicated due to $(100-h)$ coeff. Perhaps can approx. better though?

1c. you reach the correct conclusion but the distribution and the associated CI is incorrect. We have to use the CLT and find the limiting distribution of the sample mean!

This is the set of values b for which we couldn't reject $H_0: \mu = b$ at the 5% level given the sample. Since the interval contains 0, the data do not suggest that $E[x_t] = 0$.

d) You could use OLS to estimate the parameters α_0 and α_1 , where $\hat{\beta} := (\hat{\alpha}_0, \hat{\alpha}_1) = \underset{\beta \in \mathbb{R}^2}{\operatorname{argmin}} (Y - X\beta)'(Y - X\beta)$

$$\text{where } Y = \begin{pmatrix} x_1 \\ \vdots \\ x_T \end{pmatrix} \quad X = \begin{pmatrix} 1 & x_0 \\ \vdots & \vdots \\ 1 & x_{T-1} \end{pmatrix}, \quad \beta = (\beta_1, \beta_2)'$$

with solution $(X'X)^{-1} X'Y$.

1d. ok, correct

To estimate σ^2 , we could use the residual variance and let $\hat{\sigma}^2 = \frac{1}{T-2} (Y - X\hat{\beta})'(Y - X\hat{\beta})$.

These estimators are consistent (demonstrable by mds LLN) and $\hat{\beta}$ is asymptotically normal (by mds CLT).

(Are there asymptotic efficiency results here? I don't think Gauss-Markov applies since ε_t is not exogenous.)

2. $z_t = \mu + z_{t-1} + e_t$; $e_t \stackrel{iid}{\sim} N(0, \sigma^2)$. Fit $z_t = \delta t + u_t$, with $\hat{\delta}_T$.

a) $\hat{\delta}_T := \underset{d \in \mathbb{R}}{\operatorname{argmin}} \sum_{t=1}^T (z_t - dt)^2$, for which the FOC $\frac{\partial L}{\partial d} = 0$ requires

$$-2 \sum (z_t - dt) = 0 \Rightarrow \sum z_t t = \hat{\delta}_T \sum t^2$$

$$\therefore \hat{\delta}_T = \frac{\sum z_t t}{\sum t^2}$$

substituting in the true model, $\hat{\delta}_T = \frac{\sum (\mu + z_{t-1} + e_t) t}{\sum t^2}$

The true model is an AR(1) unit root process, which can be decomposed into the sum of an initial value, a deterministic trend, and a stochastic trend (in this case, a random walk).

To see why, note that $z_t = \mu + (\mu + z_{t-2} + e_{t-1}) + e_t$
 $= z_{t-2} + 2\mu + (e_{t-1} + e_t)$
 $= \dots$
 $= z_0 + t\mu + \sum_{i=1}^t e_i$

So, $\hat{\delta}_T = \frac{\sum_{t=1}^T (z_0 + t\mu + \sum_{i=1}^t e_i)t}{\sum t^2}$ **2a. ok, correct!**

$$= \frac{1}{\sum t^2} (z_0 \sum t + \mu \sum t^2 + \sum v_t) = \mu + \frac{z_0 \sum t + \sum v_t}{\sum t^2}$$

b) $E[v_t] = t E[\sum_{j=1}^t e_j] = t \sum E[e_j]$ by linearity $= \mu + \sum v_t / \sum t^2$ as $z_0=0$
 $= 0$ since e_j is mean zero.

$$\text{var}(v_t) = t^2 \text{var}(\sum e_j) = t^2 \sum \text{var}(e_j) \text{ as } e_i \perp e_j \text{ if } i \neq j$$

$$= t^3 \sigma^2$$

2b. yes, well done!

$$\text{cov}(v_t, v_s) = E[v_t v_s] \text{ since both mean zero}$$

$$= st E[\sum_{j=1}^t e_j \sum_{i=1}^s e_i]$$

and given $e_i \perp e_j$ $i \neq j$, the expectation of the product is nonzero only where $i=j$.

$$= st E[\sum \sum e_j e_i]$$

$$= st \sum \sum E[e_j e_i]$$

$$= st \cdot \min\{s, t\} \cdot \sigma^2 \text{ since } E[e_i^2] = \sigma^2$$

c) $E[\hat{\delta}_T] = \mu + E[\sum v_t / \sum t^2] = \mu + \frac{1}{\sum t^2} \sum E[v_t] = \mu + \frac{1}{\sum t^2} \cdot 0 = \mu$

$$\text{var}(\hat{\delta}_T) = \frac{1}{(\sum t^2)^2} \text{var}(\sum v_t) \text{ since } \mu \text{ is non-random}$$

Let's focus on $S := \text{var}(\sum v_t) = \sum_{t=1}^T \sum_{s=1}^T \text{cov}(v_t, v_s)$.

$$= \sum_{t=1}^T \text{var}(v_t) + 2 \sum_{t=1}^{T-1} \sum_{s=t+1}^T \text{cov}(v_t, v_s)$$

$$= \sum_{t=1}^T t^3 \sigma^2 + 2 \sum_{t=1}^{T-1} \sum_{s=t+1}^T s t^2 \sigma^2 \quad \text{Since } t=s-1 \text{ so it's always the min}$$

$$= \sigma^2 \left(\sum_{t=1}^T t^3 + 2 \sum_{t=1}^{T-1} t^2 \sum_{s=t+1}^T s \right)$$

Note that $\sum_{s=t+1}^T s = \sum_{s=1}^T s - \sum_{s=1}^t s = \frac{1}{2} (T(T+1) - t(t+1))$

$$\text{So then } S = \sigma^2 \left(\sum_{t=1}^T t^3 + \sum_{t=1}^{T-1} t^2 (T(T+1) - t(t+1)) \right)$$

We can safely change the ^{upper} limit on the second sum to T rather than $T-1$ since then the s part $= 0$, as $t=T$.

Thus $\text{var}(\hat{\delta}_T)$ takes the required form, since it $= \frac{S}{(\sum_{t=1}^T t^2)^2}$.

d) It will be useful to note that $\int_0^1 r^k dr = \frac{1}{k+1}$.

Scale the numerator and denominator of $\text{var}(\hat{\delta}_T)$ by T^{-6} :

$$\text{var}(\hat{\delta}_T) = \frac{\sigma^2}{\left(\frac{1}{T} \sum_{t=1}^T (t/T)^2 \right)^2} \cdot \left[\underbrace{\frac{1}{T^2} \cdot \frac{1}{T} \sum_{t=1}^T (t/T)^3}_{\rightarrow \frac{1}{T^2} \cdot \frac{1}{4}} + \frac{1}{T} \cdot \frac{1}{T} \sum_{t=1}^T (t/T)^2 \left(\frac{T}{T} \left(\frac{T}{T} + \frac{1}{T} \right) - \frac{t}{T} \left(\frac{t}{T} + \frac{1}{T} \right) \right) \right]$$

denominator $\rightarrow \left(\frac{1}{3} \right)^2 = \frac{1}{9}$ $= \frac{1}{T} \sum_{t=1}^T (t/T)^3 \left(1 + \frac{1}{T} - \frac{t}{T} \left(\frac{t}{T} + \frac{1}{T} \right) \right)$

Considering $1 + \frac{1}{T} - \frac{t}{T} \left(\frac{t}{T} + \frac{1}{T} \right)$, notice that each term here is at most 1, since $t \in [1, T]$. As $T \rightarrow \infty$, $\frac{1}{T} \rightarrow 0$, so this expression $\rightarrow 1 - (t/T)^2 \in [0, 1]$. So ^{the second sum} it can be treated as $\sum_{t=1}^T (t/T)^2 (1 - (t/T)^2) = \sum_{t=1}^T (t/T)^2 - (t/T)^4$.

$$\text{Therefore } \text{var}(\hat{\delta}_T) \sim 9 \sigma^2 \cdot \left[\frac{1}{T^2} \cdot \frac{1}{4} + \frac{1}{T} \left(\frac{1}{3} - \frac{1}{5} \right) \right]$$

and since $T^{-2} \rightarrow 0$ quickly, $\sim 9 \sigma^2 \cdot \frac{2}{15} \cdot \frac{1}{T} = \sigma^2 \frac{6}{5} \cdot \frac{1}{T} \rightarrow 0$ at rate $\frac{1}{T}$

For $\text{Var}(\sqrt{T} \hat{\delta}_T)$, notice that we're scaling up exactly by the square root of the rate at which $\text{var}(\hat{\delta}_T)$ shrinks, so the limiting variance will now be a constant.

Specifically, $\text{var}(\sqrt{T} \hat{\delta}_T) = T \text{var}(\hat{\delta}_T) \approx 1.2 \sigma^2$, as $T \rightarrow \infty$

- e) Yes. This is quite easy to see with a mean square argument.
- Convergence in mean square implies convergence in probability, by Chebyshev's inequality.
 - $E[(\hat{\delta}_T - \delta_T)^2] = \text{var}(\hat{\delta}_T)$ since $\hat{\delta}_T$ is unbiased as shown (c).
 - $\text{var}(\hat{\delta}_T) \rightarrow 0$ as $T \rightarrow \infty$.
 - Therefore, $\hat{\delta}_T \xrightarrow{P} \delta_T$.

2c, 2d: I'm not sure whether this is an issue at my end but I cannot see your answers for these 2 sub-questions! I see in 2c. your calculation for the expectation. After that, I direct see your answer for the variance of delta hat which is for 2d. Perhaps an error in uploading?

2e. yes, correct! note that one would have hoped that the estimator converged to zero in probability since the true process is not a linear trend